Dylan Edwards

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EXPERIENCE

Robo-Touch Lab, University of Illinois Urbana-Champaign Champaign, IL

Research Assistant *Aug. 2025 – Present*

- Designed and built a **19-DoF dexterous robotic hand** from scratch with direct-drive worm-gear joints, including selection of the motors, motor drivers, and ADCs
- Engineered a custom PCB that mounts on the back of the palm and drives all 19 joints, and implemented the embedded firmware and tuned the joint motion controllers
- Trained reinforcement learning policies in simulation for manipulation tasks and transferred them to the physical hand
- Evaluated joint accuracy, power grasp force, repeatability, and grasping capability using a grasp taxonomy
- Modeled hand kinematics in simulation to assess manipulability and reachable workspace volume

US Army Construction Engineering Research Laboratory (CERL) Champaign, IL

Undergraduate Research Assistant *Jan. 2024 – Jul. 2024*

- Developed a waterproof electronics enclosure integrating custom-designed PCBs, cabling, and housing
- Created an adjustable belt-tensioning module adaptable to multiple applications
- Documented all internal and external connections in a system-level component diagram to support integration

Siebel Center for Design Champaign, IL

Shop Attendant *Aug. 2023 – Jan. 2024*

- Operated CNC, waterjet, and laser-cutting equipment; performed soldering and hands-on fabrication
- Instructed users on CNC and waterjet operation and file setup, and advised on manufacturing process selection

Summer Intern *Jun. 2023 – Aug. 2023*

- Produced manufacturing jigs that improved process consistency and accuracy
- Redesigned and implemented a workspace layout using human-centered design principles; presented progress to clients

PROJECTS

Pneumatic Testing Apparatus, Crandall Stats & Sensors *Senior Design*

- Created modular PLC-based software architecture for an automated pneumatic quality-control test system
- Evaluated advanced flow sensors to extend the system's measurement capabilities

Thermal Eye, Mottier Innovation Challenge *2nd Place*

- Integrated an off-the-shelf thermal camera with a custom display mount for an NFPA-standard firefighter helmet, enabling hands-free scanning for danger points in burning structures
- Pitched the design to a judging panel of department members and industry professionals

SKILLS

CAD: SolidWorks, Fusion 360, Inventor, Onshape
Prototyping & Fabrication: FDM 3D printing, CNC machining, waterjet cutting, laser cutting, silicone molding, soldering
Electronics & Controls: PCB design (KiCad), DC motor control, sensor integration, embedded firmware
Software: C++, Python, ROS, PLC programming, reinforcement learning (sim-to-real)

EDUCATION

University of Illinois Urbana-Champaign May 2025

B.S. in Systems Engineering and Design, Minor in Computer Science

LEADERSHIP

Illini VEX Robotics Champaign, IL

Mechanical Team Lead / Mentor *Oct. 2021 – Dec. 2024*

- Led and mentored a 15+ member team through the design and build of 2 competition robots
- Introduced design techniques that improved team workflow and build quality

Outreach Director *Aug. 2023 – May 2024*

- Organized collegiate and high school tournaments and ran CAD challenges with other student organizations